



ORACLE

Oracle Integration 3

Customer-Managed Disaster Recovery
Solution

Customer: Qatar Foundation

Presented by: Oracle Customer Success Services (CSS)

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Executive Summary

Customer-Managed DR for Qatar Foundation's OIC 3 environment using OCI Full Stack Disaster Recovery (FSDR) with custom scripts. All switchover/failover/failback operations orchestrated via FSDR DR Plans.



Integrations

FSDR Orchestrated
REST API + DNS

Supported



File Server

FSDR Orchestrated
sFTP + LB + DNS

Supported



Visual Builder

FSDR Orchestrated
BYODB + ATP DG

Supported



Processes (OPA)

Alternate Solution
REST API + External DB

Workaround

Topology: Active-Passive | **Primary:** Frankfurt (FRA) | **DR:** Amsterdam (AMS) | **Orchestration:** OCI Full Stack DR (FSDR)

DR Scope — Customer-Managed Solution



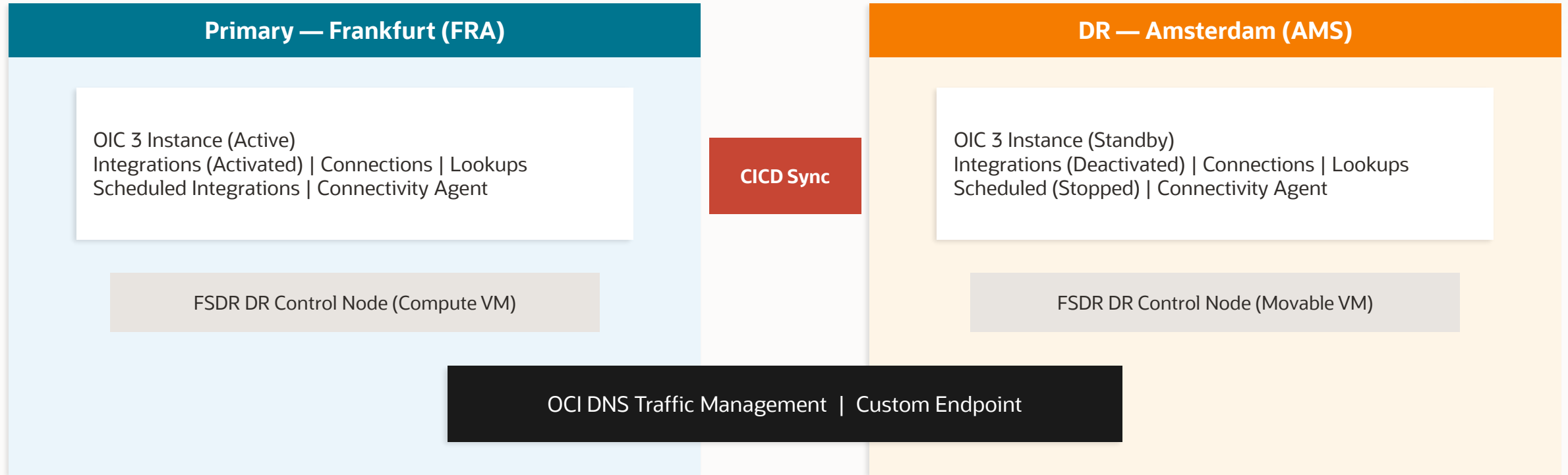
OIC 3 Component	DR Support Status	DR Method	Orchestration
Integrations	✓ Supported	REST API export/import + CICD + DNS switch	FSDR + Custom Scripts
File Server	✓ Supported	sFTP sync + Load Balancer + DNS switch	FSDR + Custom Scripts
Visual Builder (VBCS)	✓ Supported	BYODB (ATP Data Guard) + DNS switch	FSDR + Custom Scripts
Process Automation (OPA)	⚠ Workaround	REST API sync + External DB for runtime	FSDR + Custom Scripts
B2B	✓ Supported	Use Oracle-Managed DR (not customer-managed)	Oracle-Managed only
RPA	✗ Not Supported	No official DR solution available	N/A

All switchover/failover/failback operations orchestrated via OCI Full Stack DR (FSDR) with custom scripts — extended to cover all components.



Integrations

Integrations — DR Architecture



Metadata sync via OCI Object Storage. FSDR DR Plans execute custom scripts for activation, deactivation, and DNS updates during switchover/failover.

Integrations — DR Approach



What is Synced

Design-time metadata: connections, integrations, lookups, libraries, projects, packages. Exported via REST API to OCI Object Storage, imported to standby via CICD pipeline.

Custom Endpoint

Customer-managed custom endpoint configured on both instances. DNS CNAME updated by FSDR custom script during switchover/failover.

Connectivity Agent

Separate agents provisioned in each region (FRA and AMS). FSDR DR Control Node (movable compute) hosts scripts and moves between regions during DR operations.

Scheduled Integrations

Kept deactivated on standby. FSDR custom script activates/deactivates integrations during DR operations. Scheduled parameters synced periodically via REST API.

In-Flight Transactions

Cannot be failed over. Must be retried after failover. Design integrations with idempotent patterns, correlation IDs, and retry logic.

Monitoring / History

Runtime monitoring data is not replicated. Preserved via Activity Stream Logs → OCI Logging → Log Analytics (see next slide).

Integrations — Key Considerations

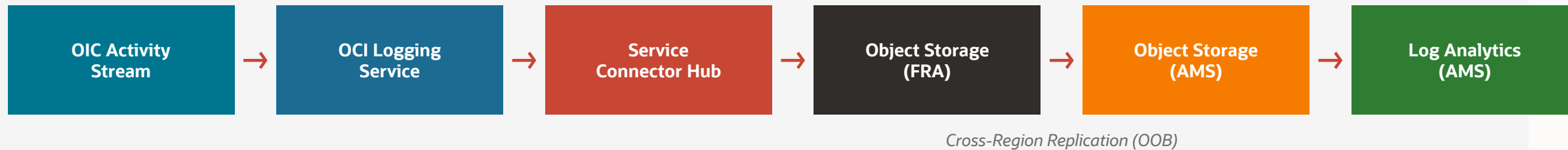


[Source: Oracle Documentation — Configuring Customer-Managed DR Solution for Oracle Integration 3](#)

-  Scheduled parameters in integrations may not be the latest in the secondary instance — depends on sync frequency. Frequently update scheduled parameters in the standby instance.
-  If a connectivity agent is required, provision and configure separate connectivity agents for the primary and secondary Oracle Integration instances.
-  When the primary instance is active, stop or deactivate scheduled flows and integrations with polling triggers in the secondary instance. After failover, activate these artifacts in the secondary instance.
-  Ensure source applications and end users use the customer-managed custom endpoint to access Oracle Integration. Administrators must use original instance hostnames to manage instances.
-  Failover is quick and simple for non-scheduled and non-polling integration flows. Scheduled flows and integrations with polling triggers require special handling.

Integrations — Activity Stream Logs (Cross-Region)

OIC runtime monitoring data is not replicated during DR. Activity stream logs (at Production tracing) capture execution metadata and fault details only — no payload data. The following pipeline replicates these logs cross-region.



- 1. Enable OCI Logging**
Enable OIC activity stream logs in OCI Logging Service for primary (FRA) instance. Configure log categories: integration execution, connection tests, deployment events, errors.
- 2. Create Object Storage Buckets**
Create OCI Object Storage buckets in FRA and AMS. Enable versioning, configure lifecycle policies (90-day retention), enable cross-region replication FRA → AMS.
- 3. Configure Service Connector Hub**
Create a Service Connector in FRA: Source = OCI Logging (OIC activity stream log group), Target = Object Storage bucket (FRA). This bridges Logging to Object Storage automatically.
- 4. Configure Log Analytics (AMS)**
Setup OCI Log Analytics workspace in AMS. Configure Object Collection Rule to ingest logs from AMS Object Storage bucket. Create dashboards and alerts.
- 5. Post-Failover**
After failover, historical logs available in AMS Log Analytics. New DR OIC instance in AMS: enable OCI Logging → Service Connector Hub → AMS bucket → same Log Analytics workspace.

Integrations — What's Not Replicated to DR

The following data is per-instance and will NOT be available on the DR instance after switchover/failover. These are accepted trade-offs or require additional customization.

Not Possible to Replicate — Handled Manually

In-flight transactions	Lost on failover. Must be retried. Design with idempotent patterns and correlation IDs.
Faulted instance resubmit queue	Pending resubmits on primary are lost. Manual re-trigger from source systems required.
OIC-internal activity stream	The OIC Monitoring UI history is per-instance. OCI Logging export is the DR copy — but it is a subset, not the full UI view.
Payload data (at Production tracing)	Production tracing does not log payloads. Only execution metadata and fault details are captured in Activity Stream.

Requires Further Customization (Not In Scope)

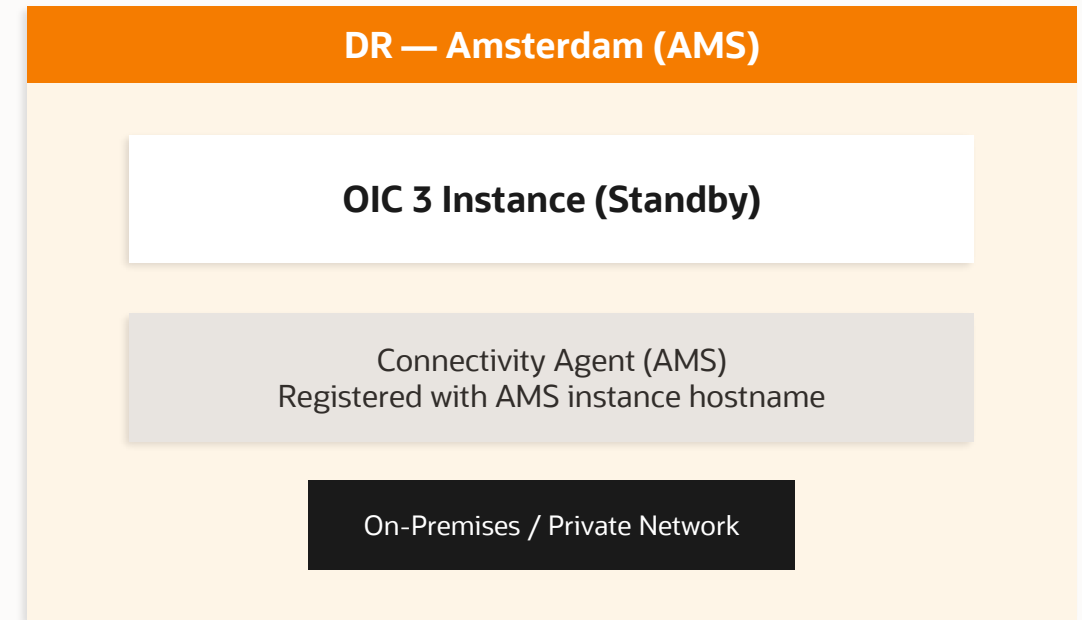
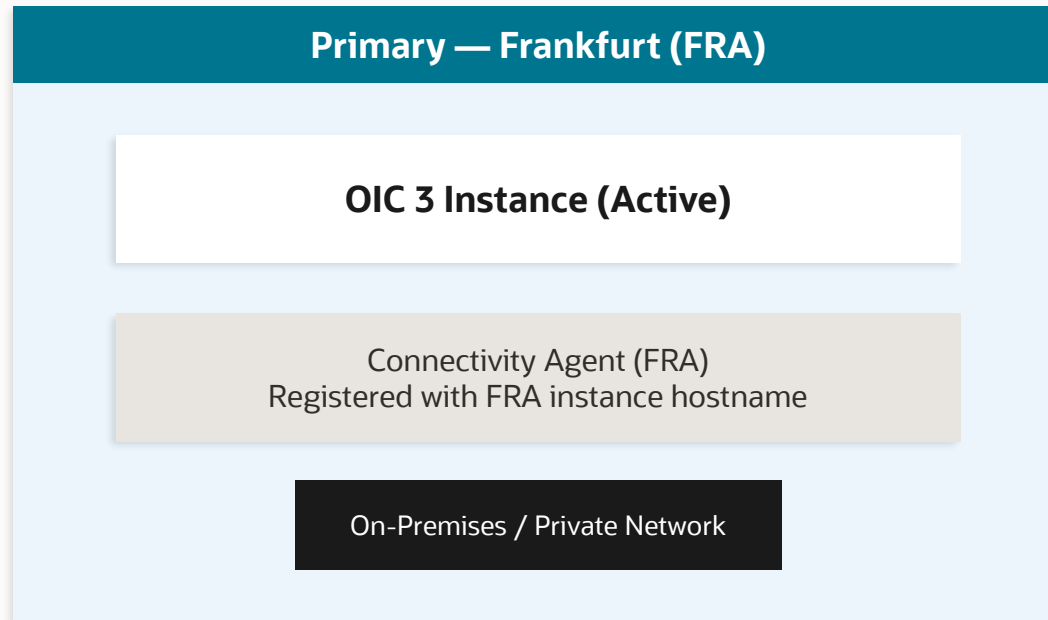
Payload-level audit logging	Add Logger action inside critical integrations to push business fields to Activity Stream (works at Production tracing). Or add in-flow DB write to external ATP for full payload capture.
Business identifier tracking data	Tracking fields are per-instance. Use Logger action to push tracking field values to Activity Stream → OCI Logging for cross-region visibility.
Monitoring dashboards / charts	Runtime metrics (message counts, success rates) rebuild automatically on DR instance once integrations run. Historical charts are lost — use OCI Logging Analytics dashboards instead.
Integration instance error details with payload	Fault messages are logged at Production tracing but failed-step payloads require Audit tracing. Use Fault Handler + Logger action to capture error payloads at Production level.

Recommendation

For critical integrations requiring payload-level DR audit: Add Logger actions to push key business fields to Activity Stream (replicated via log pipeline) + Add in-flow write to shared ATP DB (replicated via Data Guard).

Integrations — Connectivity Agent DR Approach

If integrations require access to on-premises or private network systems, a connectivity agent must be provisioned and configured separately in each region.



Setup & DR Approach

Separate Agents

Provision a dedicated agent per OIC instance. Each registers with its instance's original hostname (not the custom endpoint).

Agent Group

Use the same agent group name in both regions. Connections reference the agent group, not individual agents.

Network Access

Both agents must reach the same on-premises targets (DB, ERP, SFTP). Ensure VPN/FastConnect from both regions.

On Failover

DR agent is already running. Integrations activated on DR automatically use the local agent. No reconfiguration needed.

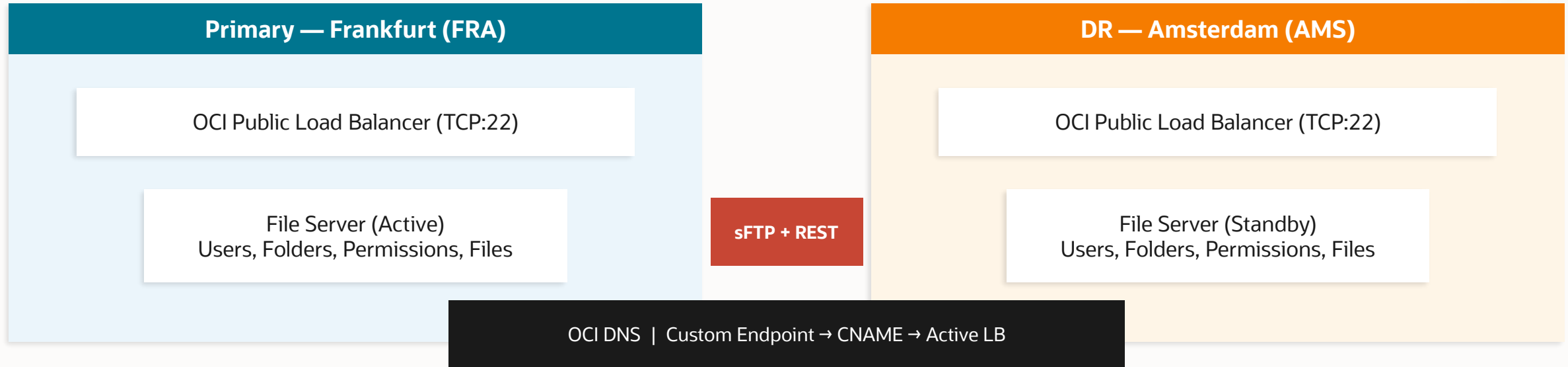
OCI Compute Hosting

Pre-provision agent VM in both regions. If using FSDR movable compute, agent VM moves with DR Control Node.



File Server

File Server — DR Architecture & Approach



Data	Sync Method
Folder structure & config	File Server REST API
File contents	sFTP rsync/mirror
Users, SSH keys, Groups	File Server REST API
Folder permissions	File Server REST API
SSL certificates	Manual upload
SFTP configuration	File Server REST API

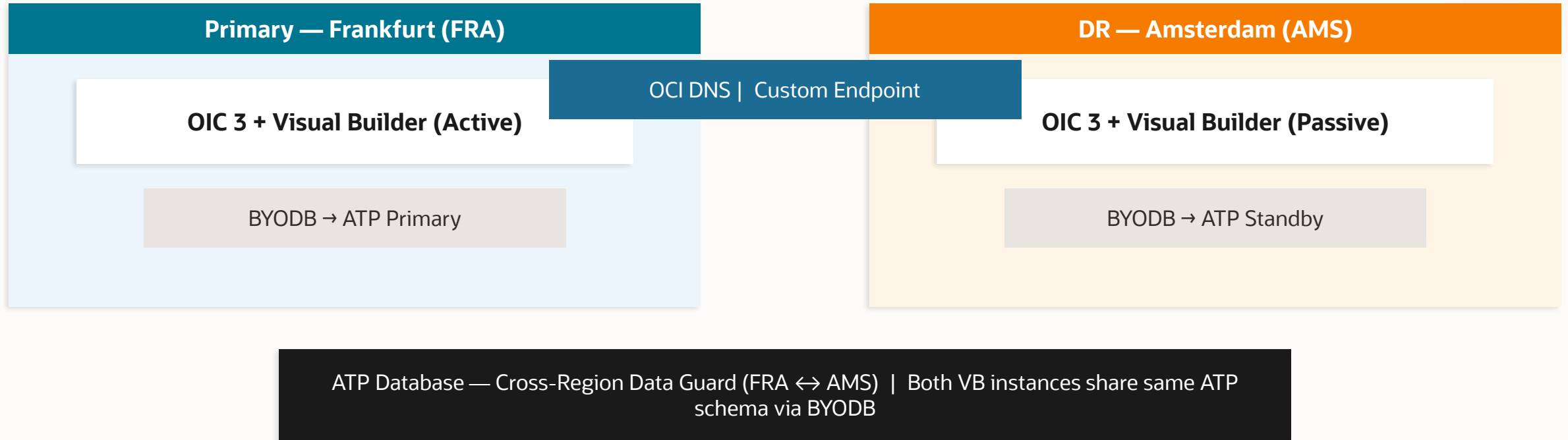
Key Considerations

- CI/CD must keep users, folders, and file data always synchronized between both File Server instances.
- File content sync requires sFTP protocol (no REST API for file upload/download).
- Each LB needs a separate custom hostname; a common CNAME parent endpoint routes to active LB.
- ServiceAdministrator OIC role required for REST API admin calls.



Visual Builder (VBCS)

Visual Builder — DR Architecture (BYODB)



Visual Builder —DR Approach & Key Considerations



One-Time Setup Steps

1. Provision two OIC 3 instances (VB enabled) in FRA and AMS
2. Provision ATP database with cross-region Data Guard
3. Configure BYODB on both VB instances → same ATP schema
4. Build custom ATP wallet with primary + standby hostnames
5. Configure shared custom domain on both instances
6. File SR → Oracle VB DevOps completes service-side setup
7. Align IDCS groups (use group-based access, not individual)
8. Update Tenant Settings: Integration & Process server URLs to use custom domain hostname

DR Approach — Key Advantages

No export/import needed — both instances serve same data.
No re-staging or re-publishing of apps required.
App data preserved via shared ATP (Data Guard replicated).
Runtime + history data available on DR via ATP standby.
DNS CNAME switch routes users seamlessly.

Key Considerations

- Cross-region Data Guard is the only supported setup. Does not support automatic failover — manual switchover required.
- IDCS identity domains replicated to other regions might not be in sync with DR region during failover.
- OCI Console may be inaccessible during region outage.
- BYODB is mandatory — without it, no shared data tier.
- Requires SR to Oracle VB DevOps for initial service-side DR setup.
- Configure IDCS Service Applications with group-based access (not individual users).



Process Automation (OPA)

Process Automation — DR Approach



OPA is excluded from both Oracle-Managed and Customer-Managed DR (Oracle Doc G10051-31, Feb 2026). No official DR path exists. Alternate solution proposed below.

Design-Time DR (REST API Sync)

Export/import process apps (.expx) and decision apps via OPA REST APIs.

Sync Pipeline:

1. List apps → Create snapshot → Export .expx
2. Upload to OCI Object Storage (FRA)
3. Cross-region replicate to AMS bucket
4. Import on DR OPA instance
5. Keep all apps deactivated

Import Order:

- Decision Apps first (no dependencies)
- Process Apps second (may reference decisions)
- Connectors / Credentials last

On Failover:

- Activate all process and decision apps
- Resume timer start events
- FSDR updates DNS records

Runtime / In-Flight DR (External DB) - Not In Scope

In-flight instances are LOST on failover.

No API to restore a process mid-flow.

No import API for audit/history data.

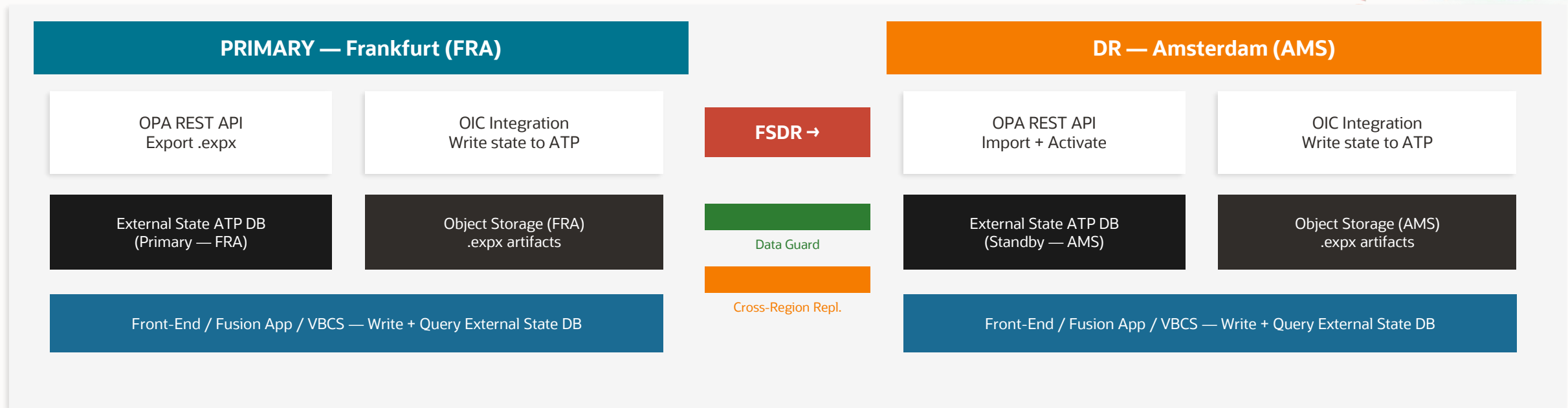
Proposed Architecture:

- External ATP database stores process runtime state at each milestone (via OIC integration or OPA service task)
- ATP replicated via Data Guard (FRA ↔ AMS)
- Front-end / Fusion App / VBCS queries this DB for process status, history, and audit

Post-Failover:

- Query external DB for incomplete instances
- Re-create from last checkpoint
- Front-end continues to show full history

Process Automation — Workaround Architecture



OPA REST API Endpoints Used

Operation	Method	Endpoint
List Process Apps	GET	/process/api/v1/design/applications
Export Process App (.expx)	GET	.../{name}/versions/{ver}/export
Import Process App	POST	/process/api/v1/design/applications
Activate / Deactivate App	PUT	.../{name}/versions/{ver}/activate .../deactivate
List / Export Decision Apps	GET	/process/api/v1/design/dmnApplications .../{name}/versions/{ver}/export
Import Decision App	POST	/process/api/v1/design/dmnApplications
Update Connector URL	PATCH	.../{name}/versions/{ver}/connectors/{id}
Resume Timer Events	POST	/process/api/v1/instances/startTimer/resume

Process Automation — Key Considerations



Key Considerations

- OPA is not officially supported in any DR model — this is an alternate/workaround approach.
- Process app export includes BPMN definitions, web forms, and decision references but NOT runtime data.
- Decision Apps and Process Apps must be exported and imported separately (different API endpoints).
- Connector base URLs must be updated on the DR instance if primary and DR OIC have different hostnames (skip if using shared custom endpoint).
- Timer/scheduled process events are paused on DR instance and must be explicitly resumed post-failover.
- External State DB adds implementation complexity but provides the only viable path for runtime continuity.
- External State DB can also serve as a centralized audit store accessible to front-end applications independent of OPA.

Proposed Custom Design Approach

1. Keep processes short-lived where possible — reduces exposure to in-flight loss.
2. External state tracking — at each key milestone, invoke an OIC integration that writes process state (instance ID, milestone, payload) to the External State ATP table.
3. Idempotent design — use correlation IDs and de-duplication checks to prevent duplicate processing on re-creation.
4. Message start events — design processes with message-based entry points that allow re-entry at known checkpoints post-failover.
5. Separation of concerns — keep business data in external DB, not solely in OPA data objects. This ensures data survives failover.
6. Front-end integration — build VBCS/Fusion screens that query External State DB directly for process status and history, decoupling from OPA workspace.



FSDR Orchestration

FSDR — Switchover & Failover Flow

All DR operations orchestrated via OCI Full Stack DR (FSDR) with custom scripts. Scripts handle metadata sync, activation/deactivation, database switchover, and DNS updates.

SWITCHOVER (Planned)

1. Deactivate scheduled/polling integrations at primary
2. Export metadata from primary → Object Storage
3. Import metadata to DR instance
4. Export and import OPA artifacts to DR instance
5. Trigger ATP Data Guard switchover (for VBCS + OPA DB)
6. Activate integrations at DR instance
7. Activate OPA apps and resume timer events
8. Update all DNS records → DR region
9. Deactivate all at primary (now standby)
10. Verify end-to-end connectivity

FAILOVER (Unplanned)

1. Primary region unreachable — trigger FSDR failover
2. FSDR moves DR Control Node VM to DR region
3. Import latest metadata snapshot to DR instance
4. Import latest OPA artifact snapshot to DR instance
5. Trigger ATP Data Guard failover (for VBCS + OPA DB)
6. Activate integrations at DR instance
7. Activate OPA apps and resume timer events
8. Update all DNS records → DR region
9. (Primary steps skipped — region inaccessible)
10. Verify end-to-end connectivity

Failback: Reverse the process — export from DR, import to primary, switchover ATP back to primary, update all DNS records back to primary region.

Thank You

